RESHAD UL KARIM

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OBJECTIVE

AI/ML engineer and published IEEE researcher who leads projects end-to-end across Generative AI and agentic LLM systems, computer vision, autonomous systems, and full-stack (MERN) development. Pre-trained transformer LLMs from scratch, architected a deployed multimodal AI agent with multi-LLM integration and an evaluation/MLOps harness, and authored peer-reviewed IEEE and conference papers; seeking to build production Generative AI, agentic, and RAG solutions.

EDUCATION

BSc. in Computer Science and Engineering, BRAC University — 09/2022 - Present
CGPA: 3.89/4.00

Higher Secondary Certificate, Notre Dame College, Dhaka — 06/2019 - 02/2022
GPA: 5.00/5.00

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, C/C++, SQL
Generative AI & LLMs	OpenAI, Claude, Mistral, OpenRouter, Prompt Engineering, RAG, Fine-tuning, RLHF
Agentic AI & NLP	Autonomous Agents, Tool Use, Planning/Memory, Embeddings, Vector Search, VLMs
ML & Computer Vision	PyTorch, Hugging Face, TensorFlow, Keras, Scikit-learn, OpenCV, MediaPipe, YOLO, SegFormer, ViT, DINOv2, SHAP
Deep Learning	CNNs, RNNs, LSTMs, Transformers, Transfer Learning
MLOps & Deployment	Training/Deployment, Experiment Tracking, Evaluation, Monitoring, Guardrails, ONNX, RunPod, VPS
Backend & Tools	REST APIs, Node/Express (MERN), MongoDB, PostgreSQL, Git, Linux
Hardware & Robotics	PCB Design, Microcontrollers, Arduino, Raspberry Pi, ROS, Sensor Integration

WORK EXPERIENCE

Undergraduate Research Assistant — 11/2025 - Ongoing
BRAC University Research Seed Grant Initiative 2025 — *Dhaka, Bangladesh*

- Contributed to DRISTEE, an AI-driven assistive navigation system for the visually impaired (computer vision, depth mapping, facial recognition, cloud processing, voice interaction), across implementation and academic writing.

Undergraduate Teaching Assistant — 02/2025 - 06/2025
BRAC University — *Dhaka, Bangladesh*

- Supported the academic progress of 120+ students across three sections by delivering tutorials on numerical algorithms, providing one-on-one guidance, and assisting with grading and assessment in collaboration with faculty.

Core Team Member – AI & Autonomous Systems — 01/2024 - 06/2025
BRAC University Mars Rover Team - MONGOL TORI — *Dhaka, Bangladesh*

- Designed high-precision electronic circuits and PCBs for the BRACU Mongol-Tori Mars Rover arm control while optimizing AI-based object detection and autonomous vision-guided systems, improving efficiency, reliability, and inference performance by up to 30%.

Contributor – Electronics and Communications — 11/2022 - 12/2023
BRAC University Rescue Rover - BRACU DICHARI — *Dhaka, Bangladesh*

- Integrated core electronics for rescue rovers (motors, drivers, microcontrollers, and rover–base station communication) and programmed microcontrollers within a ROS-based framework to improve system integration, reliability, and testing efficiency.

PUBLICATIONS

Journal

Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis
Karim, R. U., Mahdi, S., Samin, A., Zereen, A. N., Abdullah-Al-Wadud, M. M., & Uddin, J.
IEEE Access, 12 March, 2025. DOI: 10.1109/ACCESS.2025.3550577

Conference Papers

Vision-based Hand Gesture Virtual Keyboard-Mouse Framework with Bilingual Next-word Prediction
Mahdi, S., Karim, R. U., Sujat, T., Tabassum, S. M., Samin, A., & Zereen, A. N.
JCSSE 2026 – 23rd International Joint Conference on Computer Science and Software Engineering, Bangkok, Thailand, pp. 638-643. DOI: 10.1109/JCSSE68839.2026.11596610

Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification
Ferdous, T., Karim, R. U., Samin, A., Mahdi, S., Tasnim, H., Zereen, A. N.
2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE 2024), 3 March, 2025, Pages 117-122. DOI: 10.1109/ICEACE63551.2024.10898858

Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning
Ferdous, T., Karim, R. U., Samin, A., Mahdi, S., & Zereen, A. N.
2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE 2024), 3 March, 2025, Pages 117-122. DOI: 10.1109/ICEACE63551.2024.10898853

PROJECTS

Edge-Native Multimodal AI Agent for Assistive Perception 09/2024 - Present
Architected and deployed an agentic AI system with planner/router, tool use, and vector memory; multi-LLM/VLM integration (Mistral, OpenRouter) with RAG-style retrieval, a structured evaluation harness, and anti-fabrication guardrails; cut cloud vision calls 95%+.

Novel Positional Encoding for Long-Context Transformer LLMs
Led design of a novel rotary positional encoding for transformer LLMs; pre-trained 160M/420M models from scratch (PyTorch) with BF16 stability analysis.

Reinforcement Learning for Mathematical Reasoning in Low-Resource LLMs
Applied reinforcement learning to improve LLM mathematical reasoning in a low-resource language.

Leveraging Paraphrase Geometry for Model-Agnostic Embedding Enhancement
Model-agnostic method using paraphrase geometry to improve text embeddings and retrieval/vector-search quality.

CAMDet: Camera-Agnostic Metric Object Detection with Geometric Consistency
Built a monocular 3D detection framework that generalizes across cameras without retraining – DINOv2 backbone, canonical camera-space transform, joint 2D/3D detection with metric-depth estimation, geometric-consistency depth fusion, and uncertainty quantification – optimized for real-time edge inference (research, targeting CVPR/ICCV 2027).

Vision Transformer Framework for Early Detection of Autism Spectrum Disorder (ASD) in Toddlers
07/2025 - Present
Developed a ViT-based multimodal fusion framework integrating facial and behavioral datasets for scalable early screening of ASD.

Explainable Pedestrian–Vehicle Conflict-Risk Prediction in Bangladeshi Road Scenes Ongoing
Multi-modal fusion framework combining SegFormer road segmentation, YOLO13 detection, and MediaPipe pose estimation to predict pedestrian–vehicle conflicts; ensemble achieved 95.48% accuracy with SHAP-based explainability.

DRISTEE: Depth-aware Recognition and Intelligent Scene Translation for Enhanced Engagement
04/2025 - Present
Engineered a real-time navigation assistant with depth mapping, object detection, facial recognition, and voice-command NLP for visually impaired users.

WeHeal: Full-Stack Telemedicine SaaS (MERN) (live: weheal.live)
Built and deployed a full-stack telemedicine platform: REST APIs, role-based dashboards, video consultations, EMR/e-prescription, and payment integration.

Modular Vision-Based Autonomous Robot with Manipulation Capabilities (GitHub)
Built a distributed dual-ESP32 robot (WiFi TCP/IP) with ArUco marker-based point-to-point navigation and Position-Based Visual Servoing (PBVS) for autonomous pick-and-place; camera-to-body frame transforms, PID + finite-state control, and RBF-interpolated pixel-to-joint calibration.

Arduino-Based Wearable Multi-Sensor Health Monitoring & Fall Detection for Elderly Care (report)
Built a dual-Arduino wearable integrating body temperature, heart rate, SpO2, GSR stress, and MPU6050 motion sensing with GPS and GSM/Wi-Fi; delivers local (buzzer/LED) and remote (SMS with GPS location) alerts in real time on health anomalies or fall events; achieved 90% fall-detection accuracy in field tests.

Academic Success Prediction Model
Developed and evaluated multiple classification models leveraging socio-economic, demographic, and academic features to predict student outcomes as Graduate, Dropout, or Enrolled.

Assembly-Language Multi-Game Arcade Emulator (GitHub)
Built a DOS multi-game arcade (Space Car, Snake, Rapid Roll) in x86 Assembly with a unified menu launcher; used BIOS interrupts (INT 10H/16H/1AH/21H) and direct VGA memory + PC-speaker access for real-time graphics, input, timing, and sound.

CERTIFICATIONS

AI Engineer for Data Scientists Associate, DataCamp 09/2025
Credential ID: AEDS0010426564518

AI Fundamentals, DataCamp 09/2025
Credential ID: AIF0025589936888

HONORS & AWARDS

Highest Distinction, BRAC University

Vice Chancellor's List (5×) and Dean's List (5×), BRAC University 01/2023 - Present

Globally Ranked 8th, University Rover Challenge 2025, The Mars Society – Team BRACU Mongol-Tori

First Runner Up, System Analysis and Design Project, School of Data and Sciences, BRAC University Summer 2025

Second Runner Up, Robosoccer Competition TECHSPECTRA, Robotics Club of BRAC University

Third Runner Up & Grand Finalist, EXCELERATE 2025, Finance and Accounting Club of BRAC University

Semi Finalist, On-Campus Rounds, Hult Prize BRAC University 2023

Bronze Award, The Duke of Edinburgh's International Awards 05/2023 - 03/2024

VOLUNTEER EXPERIENCE

Lifetime member & Former General Secretary 02/2020 - 04/2022
Notre Dame Cultural Club *Dhaka, Bangladesh*
Organised and advised committees for national events with 5000+ attendees; drove 10x social media growth, managed cross-departmental operations, and trained succeeding executive batches in administrative and club affairs.

Secretary - Marketing, IT, Archives and Photography 12/2024 - 12/2025
BRAC University Cultural Club *Dhaka, Bangladesh*
Managed event promotions across digital platforms while leading a team of 6; drove 3x social media growth by redesigning handles and creating original design and illustrations for university-wide cultural initiatives.

REFERENCES

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